D.2.2 Pythia

For experiments on the Pythia suite, we finetune with either vanilla or myopic descent on 10M sequences of 64 tokens each subsampled from The Pile (Gao et al., 2020). Learning rates and batch sizes for each model are presented in Table 5; they are the same between vanilla and myopic descent. All other training and architectural hyperparameters are the same as those used by Biderman et al. (2023).

Model	Learning rate	Batch size
Pythia 14M	4.0×10^{-4}	512
Pythia 31M	4.0×10^{-4}	512
Pythia 70M	1.2×10^{-4}	512
Pythia 160M	1.2×10^{-4}	512
Pythia 410M	1.2×10^{-4}	256
Pythia 1B	1.2×10^{-4}	128
Pythia 1.4B	1.2×10^{-4}	128
Pythia 2.8B	8.0×10^{-5}	64

Table 5: Pythia suite hyperparameters for finetuning. Batch size is measured in sequences of 64 tokens each.

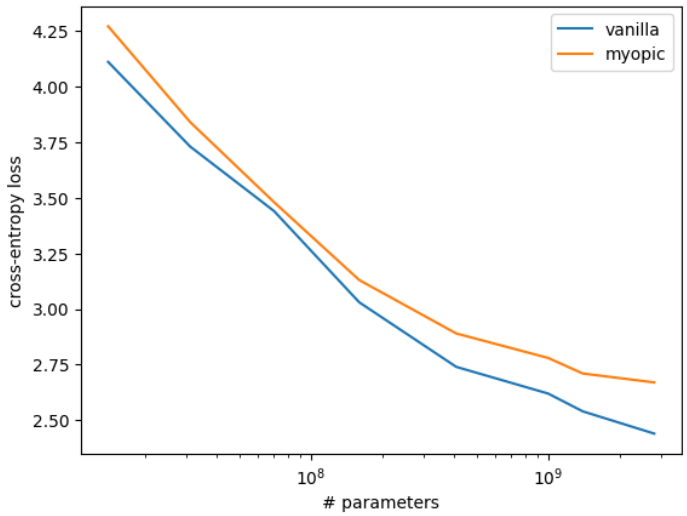


Figure 7: Cross-entropy loss of Pythia models fine-tuned on The Pile dataset using vanilla and myopic gradient descent. Starting from the 70M model, we see that the gap increases with parameter count.

D.3 Multiplication

In addition to the natural language experiments, we also measure the myopia gap on the task of natural number multiplication. We use the same GPT2-small architecture as in the natural language experiments starting from random initialization; see Table 4. See Figure 8 for an example input sequence.

```
3 7 0 0 * 5 4 0 0 = 5 8 2 3 0 0 0 0
```

Figure 8: Example multiplication input sequence.

	1	2	3	4	5	6	7	8
1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
4	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99
5	1.00	1.00	1.00	1.00	0.98	0.92	0.21	0.10
6	1.00	1.00	0.99	0.99	0.59	0.16	0.02	0.02
7	1.00	1.00	1.00	0.96	0.16	0.03	0.00	0.00
8	1.00	1.00	0.99	0.51	0.06	0.01	0.00	0.00

Vanilla multiplication accuracy

	1	2	3	4	5	6	7	8
1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
4	1.00	1.00	1.00	1.00	0.70	0.50	0.34	0.26
5	1.00	1.00	1.00	0.99	0.45	0.16	0.07	0.03
6	1.00	1.00	1.00	0.96	0.27	0.05	0.01	0.00
7	1.00	1.00	1.00	0.60	0.11	0.02	0.00	0.00
8	1.00	1.00	1.00	0.41	0.05	0.00	0.00	0.00

Myopic multiplication accuracy

	1	2	3	4	5	6	7	8
1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-0.00
4	0.00	0.00	0.00	0.00	0.30	0.50	0.65	0.73
5	0.00	0.00	-0.00	0.01	0.53	0.77	0.13	0.07
6	0.00	0.00	-0.01	0.03	0.32	0.11	0.01	0.02
7	0.00	0.00	-0.00	0.36	0.05	0.01	-0.00	0.00
8	0.00	0.00	-0.01	0.10	0.01	0.01	0.00	0.00

Vanilla-myopic accuracy gap

Figure 9: Accuracy of vanilla and myopic transformers trained on multiplication of up to 8-digit inputs. Row and columns correspond to the number of digits in the first and second multiplicands, respectively.

We use several formatting tricks found by Shen et al. (2023) to improve performance on the multiplication task:

- Characters are delimited by spaces, such that each digit is tokenized into a separate token.
- All numbers are written in the reverse of the standard order, i.e. such that the least significant digits come first.
- All inputs are zero-padded to the same length.

Note for each multiplicand we first sample the number of digits $d \sim \text{Unif}(n)$ uniformly in some range n, then uniformly sample a natural number $x \sim \text{Unif}(10^d - 1)$ with no more than d digits. This distribution allocates increased weight to smaller numbers, and was found to result in superior performance.

We train both vanilla and myopic transformers on one epoch of 10M examples with no more than 8 digits, then measure $0/1$ accuracy (that is, the model is provided with the an input sample up to the '=' token, and scored 1 if it completes the rest of the sequence exactly correctly and 0 otherwise) on 1024 independent random validation examples for each of the 8×8 possible pairs of digit counts for the two multiplicands. See Figure 9.

D.3.1 Filler tokens

We further hypothesize that, as in Pfau et al. (2024), the vanilla transformer may learn to perform computation even on forward passes corresponding to filler tokens, thus attaining better performance when trained on examples zero-padded to longer lengths. We expect that myopic transformers, on the other hand, are not incentivized to do this, since this extra computation holds no relevance towards the immediate task of predicting the filler zero token. To test this hypothesis, we train vanilla and myopic transformers on each of two different multiplication datasets:

1. Both multiplicands have at most 5 digits, and are zero-padded to exactly 5 digits.
2. Both multiplicands have at most 5 digits, and are zero-padded to exactly 10 digits.

Again, all training runs consist of one epoch of 10M examples.

See Figure 10 for results. Note that the vanilla transformer indeed performs better when trained and evaluated on input sequences zero-padded to a longer length. However, the myopic transformer performs substantially worse with increased zero-padding. Our explanation is that, not only does the myopic transformer not learn to perform intermediate tokens during zero-token forward passes, the increased input length makes it more difficult for the attention mechanism to correctly attend to the relevant tokens.

D.4 Gradient angles

Using the publicly available training checkpoints for Pythia-410M (Biderman et al., 2023), we measure the sizes of both the myopic component and the future component of the gradient of the loss w/r/t the parameters over the course of training. (Note that the future component is the difference between the total vanilla gradient and the myopic gradient.) We also measure the cosine similarity between the myopic and future components. See Figure 11. One observation is that the norm of the myopic gradient is consistently larger than that of the future gradient—thus, training is dominated by the effect of each forward pass's parameters on the immediate next-token prediction.

	1	2	3	4	5
1	1.00	1.00	1.00	1.00	1.00
2	1.00	1.00	1.00	1.00	1.00
3	1.00	1.00	1.00	1.00	1.00
4	1.00	1.00	1.00	1.00	1.00
5	1.00	1.00	1.00	1.00	0.40

Vanilla accuracy, padded to length 5

	1	2	3	4	5
1	1.00	1.00	1.00	1.00	1.00
2	1.00	1.00	1.00	1.00	1.00
3	1.00	1.00	1.00	1.00	1.00
4	1.00	1.00	1.00	1.00	1.00
5	1.00	1.00	1.00	1.00	0.99

Vanilla accuracy, padded to length 10

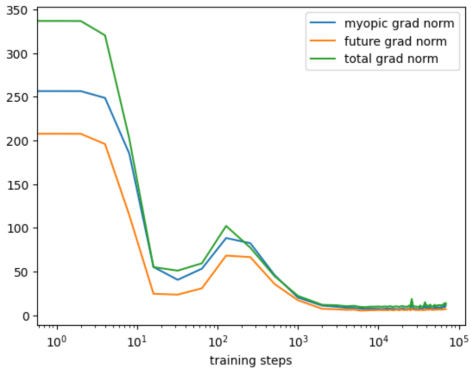
	1	2	3	4	5
1	1.00	1.00	1.00	1.00	1.00
2	1.00	1.00	1.00	1.00	1.00
3	1.00	1.00	1.00	1.00	1.00
4	1.00	1.00	1.00	0.98	0.79
5	1.00	1.00	1.00	0.82	0.39

Myopic accuracy, padded to length 5

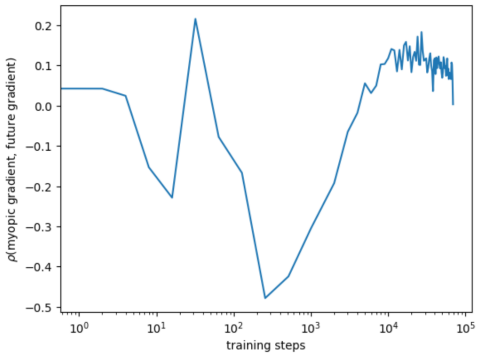
	1	2	3	4	5
1	1.00	1.00	1.00	1.00	1.00
2	1.00	1.00	1.00	1.00	1.00
3	1.00	1.00	1.00	1.00	0.99
4	1.00	1.00	0.99	0.43	0.24
5	1.00	1.00	0.97	0.22	0.05

Myopic accuracy, padded to length 10

Figure 10: Multiplication accuracy of GPT-2 with either vanilla or myopic training and with input multiplicands zero-padded to either length 5 or 10. Row and columns correspond to the number of digits in the first and second multiplicands, respectively. Observe that padding improves performance of the vanilla model, but decreases performance of the myopic model.



Norms of myopic, future, and total gradients



Cosine similarity between myopic and future gradient

Figure 11: Myopic and future gradients of Pythia-410M during training.